

4-4-1: Introduction to VLSM

At the end of this episode, I will be able to:

1. Identify and explain the purpose of variable-length subnet masking, given a scenario.

Learner Objective: *Identify and explain the purpose of Variable-Length Subnet Mask or VLSM, given a scenario*

Description: In this episode, the learner will examine the role Variable-Length Subnet Mask performs in IP subnetting. We will explore VLSM, how it works, and how it can optimize IP address allocation within IP-based networks.

- What is variable length subnet mask (VLSM)?
 - o VLSM divides an IP network into subnetworks of different sizes for more efficient IP address allocation.
 - o It allows granular address allocation based on subnet needs, optimizing IP space usage.
- How does VLSM work in IP subnetting?
 - o VLSM assigns different subnet masks to subnets, allowing variable host numbers in each.
 - o It involves dividing larger subnets into smaller ones, each with a unique mask.
 - o Careful planning is needed to prevent address conflicts and ensure proper allocation.
- Example 1 - Subnetting a 192.168.0.0/24 network
 - o Scenario: You have a network with an IP range of 192.168.0.0/24. This network needs to be divided into three subnets - HR needs 50 hosts, Sales needs 30 hosts, and IT needs 10 hosts.
 - o Subnetting:
 - ♣ For 50 hosts
 - ♣ You need at least 6 bits for hosts ($2^6 = 64$, $64 - 2 = 62$ usable addresses)
 - ♣ The subnet mask will be /26
 - ♣ Subnet 1: 192.168.0.0/26
 - ♣ For 30 hosts
 - ♣ 5 bits are needed ($2^5 = 32$, $32 - 2 = 30$ usable addresses)
 - ♣ The subnet mask will be /27
 - ♣ Subnet 2: 192.168.0.64/27
 - ♣ For 10 hosts
 - ♣ 4 bits are sufficient ($2^4 = 16$, $16 - 2 = 14$ usable addresses)
 - ♣ The subnet mask will be /28.
 - ♣ Subnet 3: 192.168.0.96/28
- Subnet details
 - o Subnet 1 - 192.168.0.0/26
 - ♣ Network ID - 192.168.0.0
 - ♣ Broadcast Address - 192.168.0.63
 - ♣ Usable IP Range - 192.168.0.1 to 192.168.0.62
 - ♣ IP Range - 192.168.0.0 to 192.168.0.63
 - ♣ Number of Usable Hosts - 62
 - o Subnet 2 - 192.168.0.64/27
 - ♣ Network ID - 192.168.0.64
 - ♣ Broadcast Address - 192.168.0.95
 - ♣ Usable IP Range - 192.168.0.65 to 192.168.0.94
 - ♣ IP Range - 192.168.0.64 to 192.168.0.95
 - ♣ Number of Usable Hosts - 30
 - o Subnet 3 - 192.168.0.96/28
 - ♣

- Network ID - 192.168.0.96

♠ Broadcast Address - 192.168.0.111

♠ Usable IP Range - 192.168.0.97 to 192.168.0.110

♠ IP Range - 192.168.0.96 to 192.168.0.111

♠ Number of Usable Hosts - 14

- For quick reference

- o 1 bit added to the node portion = 2 networks/126 nodes (remember minus the network ID and broadcast)
 - o 2 bits added = 4 networks/62 nodes
 - o 3 bits added = 8 networks/30 nodes
 - o 4 bits added = 16 networks/14 nodes
 - o 5 bits added = 32 networks/6 nodes
 - o 6 bits added = 64 networks/2 nodes
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- Additional Resources:

- o If applicable